

Joshua Sternadel

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jsternadel.github.io/portfolio/

Principal Systems & Software Engineer

Core Competencies: Full-Stack Architecture (Angular/Node/PHP/.NET) | Infrastructure Automation (Ansible, AAP) | Resilient Systems Design

Enterprise Scale: 13+ Years Systems Engineering | Core Technical Advisor and Anchor for Fortune 500 Telecom Mergers

Key Achievements

- **Designated Mission-Critical Personnel:** Awarded a major retention bonus during the Sprint/T-Mobile merger to personally stabilize infrastructure and drive technical architecture integration throughout the corporate transition.
- **U.S. Patent Author (#10,366,069):** Co-authored a patent for systems and methods for database management and administration, engineering a custom SQL natural sorting algorithm to optimize data retrieval pipelines.
- **Engineered a Custom Data Framework:** Authored a data-agnostic PHP framework using reflection and annotations to unify SQL, NoSQL, JSON, and XML into a single ORM, reducing boilerplate by 10x for new services.
- **Edge Algorithmic Processing:** Developed a local signal-filtering solution for intermodal logistics on early Android hardware, parsing raw NMEA 0183 streams and applying rate-of-acceleration modeling to eliminate coordinate jitter in high-interference rail yards.

Professional Experience

T-Mobile | Principal Solutions Architect & DevOps Engineer | 2021 to 2026

- **Architected and executed** the zero-loss migration of 10+ years of Sprint Jira Data Center metadata and device certification data to T-Mobile's Jira Cloud within 3 months, maintaining zero downtime for 3,000+ active enterprise users via a custom PHP syncing framework.
- **Engineered a custom automation framework** by authoring fluent Node.js API SDKs and Node-RED nodes integrated with Ansible to instantly capture runtime playbook errors and auto-generate structured Jira incidents.
- **Flatlined a critical vulnerability backlog** of 100+ monthly exploits across 97 hosts, solo-engineering an idempotent Ansible patch management framework managing the full

deployment lifecycle of core enterprise security agents in an a lab environment where developers retained root access.

- **Authored a serverless Angular Universal & Express application** that served as a high-visibility 5G Speed Calculator for the network team, guaranteeing instant scalability and zero baseline infrastructure overhead.
- **Architected the “Branded Experience Portal” monorepo**, engineering highly performant, T-Mobile-themed universal components optimized for modular distribution as independent corporate NPM packages.
- **Automated fleet mobilization** across 132 Spirent Landslide traffic-generation VMs, programmatically modifying Netplan configurations using defensive Ansible staging loops to eliminate remote host lockouts.

Sprint | Principal Engineer & Application Developer III | 2017 to 2021

- **Architected and engineered “Porcupine”**, a custom event-driven message bus on RabbitMQ that consumed live Jira workflow states to drive real-time automation and eliminate cross-system data lag. **Automated a legacy, 10-year-old enterprise SKU bottleneck** within 6 months by designing a Human-in-the-Loop (HITL) automation pipeline and single-handedly engineering 100% of the frontend application layer for the SKU Portal.
- **Streamlined complex retail system mappings** by integrating the SKU Portal with Porcupine, coordinating technical data inference with multi-stage TPM and reviewer approval workflows.
- **Engineered a data-agnostic PHP framework** featuring a custom hydration engine using reflection and annotations, unifying disparate data formats (SQL, NoSQL, JSON, XML) into an entity-managed ORM that reduced boilerplate by 10x.
- **Architected and validated a Dremio data lakehouse POC** to federate data layers across distributed MS SQL, MySQL, and MongoDB environments, successfully proving zero-ETL query federation across relational and NoSQL silos.
- **Built an integrated observability platform** across 45 VMs, utilizing Docker consumers to listen to .NET Core heartbeat telemetry broadcasted over Porcupine and ingest metrics into ELK, Prometheus, and Grafana with automated Rocket.Chat alerts.
- **Served as Principal Technical Advisor and Gatekeeper** to leadership, holding final engineering sign-off authority to halt high-risk deployments while scaling and managing the departmental infrastructure budget from \$300K to \$1.3M.

Sprint | Product Development Engineer II | 2013 to 2017

- **Architected and built a visual drag-and-drop prototyping platform** to simulate Out-of-Box Experiences (OOBE), engineering custom validation engines to capture real-time CPU, RAM, and battery telemetry from mobile devices.

- **Authored the foundational codebase** for the core Device Development and Certification requirements database, centralizing nationwide test-plan execution and scalable compliance tracking.
- **Served as Principal Technical Liaison to the Head of Retail Testing**, directing weekly engineering reviews to resolve complex system anomalies and establish release timelines for the nationwide SLATE ecosystem.
- **Leveraged hardware validation expertise** to inject strict performance, resource, and battery-drain constraints directly into early-stage software and application architectures.

EXPERIS (Contractor at Sprint) | Technical Support Rep II | 2012 to 2013

- **Engineered and maintained core XML execution scripts** driving the automated SLATE diagnostic framework deployed across Sprint's entire national retail footprint.
- **Optimized legacy device performance tracking applications** to ensure flawless execution of automated, on-site device validation routines for in-store service centers.

MO-KAN Container Services | System Administrator | 2011 to 2012

- **Eliminated severe physical infrastructure risks** by spearheading the migration of failing, legacy HP G3 bare-metal environments (Windows Server 2003) to a modern, high-availability architecture.
- **Brokered and negotiated a vendor partnership** to design and deploy a redundant VMware virtualized environment, integrated with an iSCSI SAN and Cisco switching infrastructure.
- **Upskilled rapidly to master virtualization technologies**, taking complete operational ownership of the new infrastructure to reduce enterprise downtime by 90% across 3 distributed corporate locations.
- **Managed end-to-end technology operations**, including disaster recovery planning, automated failover configurations, network security, and centralized domain architectures.

Additional Professional Experience

Trucking Central | Software Consultant (Contract)

- **Developed mobile supply chain solutions**, engineering early Android-based GPS applications and telematics integrations optimized for real-time intermodal logistics tracking.

Innovative Solutions | Mobile Solutions Technician

- **Engineered tactical hardware solutions**, deploying and optimizing Nextel communication devices and mobile diagnostic frameworks for municipal first responders and commercial operations.

Technical Environment Matrix

- **Patents & Core Innovation:** US Patent #10,366,069 (Systems and Methods for Database Management and Administration)
- **Architectural Disciplines:** Systems Architecture, Distributed Systems, Microservices, API Gateway Design, Platform Abstractions, SRE, DevSecOps, Enterprise Cloud Migration
- **Infrastructure & Automation:** Ansible, Ansible Automation Platform (AAP), Infrastructure as Code (IaC), CI/CD Pipelines, Node-RED, Core XML Scripting, Serverless, VMware vSphere/ESXi, iSCSI SAN Architecture
- **Data Engineering & Distributed Ecosystems:** RabbitMQ (Porcupine Message Bus), Dremio (Lakehouse Federation), Zero-ETL Query Federation, ELK Stack, Prometheus, Grafana, PostgreSQL, MS SQL, MongoDB, Rocket.Chat Integration
- **Backend Environments & Languages:** Node.js Runtime, .NET Core (2.x), C#, PHP, Native Modules, REST APIs & Custom SDK Development, Zero-Dependency Design, Custom ORM Design
- **Frontend Web Frameworks:** Angular (Pre-1.0 to v15), Vue.js 2, TypeScript, JavaScript (ES6+), Client-Side Architecture, UI/UX Performance Optimization

Community Impact & Leadership

Seek and Serve (501c3) | Technical Advisory Board Member | 2026 to Present

- **Provide architectural oversight** and technical logistics strategy to support regional community outreach expansion.

33 Meals | Technical Advisor & Infrastructure Volunteer | 2026 to Present

- **Engineered the digital architecture** and logistics pipeline for the non-profit subsidiary to systematically address regional food insecurity.
- **Established core system deployments**, managing secure domain configurations and compliance planning to ensure zero-overhead distribution operations.
- **Designed a data-driven delivery framework** applying enterprise architecture efficiency principles to scale real-world humanitarian food delivery.

Education

- **Associate of Arts (AA), Graphic Design** – Johnson County Community College